

× PROOF-OF-REASONING

● MAINNET · LIVE

ERC-8004 REFERENCE IMPL

# TuringVault

AI-powered RWA portfolio management infrastructure on Mantle — every allocation **survives adversarial challenge** before execution, anchored on chain.

ASSET CLASS

**Tokenized Treasuries**

USDT0 · mETH · USDY

CHAIN

**Mantle Mainnet**

5 verified contracts

VERIFICATION

**3-model + ERC-8004**

GLM-5 · Claude 4.6 · Gemini 3.5

WHAT MANTLE JUDGES SAID MATTERS

# The hackathon brief lists **three defining features**. We built our entire stack around them.

*"On-chain benchmarking of AI · ERC-8004 agent identity standard · radical transparency" — Mantle Turing Test 2026 brief*

## 01 On-chain benchmarking of AI

Every cycle writes 4 attestation TXs to Mantle Mainnet (proposal · validation · decisionLog · reputation). 104+ decisions logged with full reasoning hashes. The hourly cron's public GitHub Actions workflow log gives judges a verifiable artefact, not a claim.

## 02 ERC-8004 agent identity standard

Reference implementation deployed on Mantle Mainnet — non-transferable identity NFT with auto-updating tokenURI, on-chain reputation accumulating from settled outcomes, validation registry for pre-action attestation. Ready to interop with Mantle's official Agent ID issuance.

## 03 Radical transparency

Public hourly cron · IPFS-pinned reasoning · live mascot reflecting real cycle freshness · /challenge page lets visitors probe the agent's gates · /discipline page shows post-execution proof per cycle · honesty-rule documented (.kiro/steering/no-lying-about-state.md) and enforced.

*Live-stream ready: GitHub Actions workflow runs hourly, judges can drop into any past or current run.*

## WHY THIS MATTERS

# AI agents allocate real capital. Nobody can verify why.

## TODAY'S AUTONOMOUS AGENT

- Receives capital
- Black-box LLM produces decision
- Signs and broadcasts on-chain
- Reports PnL screenshot

No audit trail · No challenge · No accountability.

## TURINGVAULT

- Receives capital
- **Three independent models challenge each other**
- **Every decision hashed → IPFS → Mantle**
- **Post-execution proof verification (Discipline Layer)**

Refusal-with-proof > profitable execution without one.

For RWA allocation specifically — DAO treasuries, DeFi-native funds, compliance-conscious operators — accountability is not a feature. It's the entire reason RWA assets exist.

## OUR POSITIONING

# Not a trading bot.

## An accountable RWA portfolio manager.

## RISK-ON LEG

**mETH**

Mantle native ETH staking — real yield

## TREASURY ALLOCATION

**USDT0**

LayerZero-bridged Tether — Treasury-collateralised

## RISK-OFF FLOOR

**USDT**

Stablecoin · pre-allocation buffer

## DECISION QUALITY

**65%+**

Validator rejection rate · capital preservation

## RWA EXPOSURE

**74% NAV**

Currently in tokenized Treasuries (USDT0)

## PARSE STABILITY

**100%**

Structured-output success across 24h smoke

Compliance-conscious DAOs need an agent that can *prove its restraint*, not just its profits. Every blocked allocation has on-chain reasoning anchored to Mantle and pinned to IPFS.

## HOW A DECISION FLOWS

Every allocation passes **three independent models** and **three on-chain attestations**. Then a fourth gate verifies the execution itself.

**1. ANALYST****Z.ai GLM-5**

Proposes allocation from unified market context (CoinGecko, Nansen MCP, Hyperliquid funding, DeFiLlama TVL, Elfa social attention).

**2. VALIDATOR****Anthropic Claude 4.6**

Adversarial gate. Default REJECT. Burden of proof on the proposer. R/R  $\geq 1.5:1$  mandatory.

**3. ARBITER****Google Gemini 3.5**

Tiebreaker on disagreement. 2-of-3 consensus required to execute.

```
Hourly GitHub Actions cron (public verifiable log)
  |
  v
5 data sources: CoinGecko · Nansen MCP · Hyperliquid · DeFiLlama · Elfa social attention
  |
  v
Signal Engine: 5 structured signals (funding · on-chain flow · yield spread · liq map · social)
  |
  v
unified market context → Analyst → Validator → Arbiter
  |
  v
4 on-chain attestations: proposal · validation · decisionLog · reputation
  |
  v
RWA Allocator (Path A LLM-driven · Path B deterministic idle-parking)
```

VERIFIABLE ON-CHAIN RIGHT NOW

# Everything below is **verifiable on Mantlescan**.

**102+**

ON-CHAIN DECISIONS

**65%**

VALIDATOR BLOCK RATE

**74%**

NAV IN RWA

**100%**

PARSE SUCCESS (24H)

## SMART CONTRACTS (SOURCIFY)

Identity  
DecisionLog  
ValidationRegistry  
ReputationRegistry  
Router

0x6f86...28bD  
0x7bCd...cfbB5  
0x6841...63b6  
0xC781...6e1a  
0x8187...7001

## PUBLIC ARTIFACTS

First RWA swap (Mantlescan)  
`0x0af2336...3e09de`  
Hourly cron (GitHub Actions)  
[Agent Cycle workflow](#) · [public log](#)  
Reasoning blobs (IPFS via Pinata)  
[CHALLENGE-\\*](#) · [PoR-\\*](#) · [AgentCard-v\\*](#)  
Live social signal (Elfa)  
[/api/elfa-snapshot](#) · [v2 REST](#)

Operator EOA `0xDC78...fb5a` · Mantle chain 5000 · all contracts verified

## RADICAL TRANSPARENCY

# We hold ourselves **to the same standard** we ask of the agent.

A project that promises Proof-of-Reasoning cannot afford fabricated metrics. Every claim on the dashboard is traceable to a contract read, a settled outcome, or a public file.

|      |                                      |                         |
|------|--------------------------------------|-------------------------|
| LIVE | AI consensus pipeline                | 102 on-chain TXs        |
| LIVE | ERC-8004 identity + reputation       | 5 verified contracts    |
| LIVE | Hourly cron via GitHub Actions       | public workflow log     |
| LIVE | Discipline Layer post-execution gate | 3 gates fire each cycle |
| LIVE | RWA allocation to USDT0              | first swap 0x0af2336... |

Source-of-truth: `.kiro/steering/no-lying-about-state.md`

# Infrastructure for **DAO-grade RWA allocation**, built natively on Mantle's primitives.

*AI x RWA Track Path A scoring (40%): "Completeness of asset tokenization flow + innovation of technical approach". TuringVault is the rare submission that ships all three defining features end-to-end.*

## METH

### Native ETH staking

Risk-on RWA leg with real on-chain yield, sourced from Mantle's own LST

## USDT0

### LayerZero omnichain Tether

\$1.1M Merchant Moe LB v2.2 pool — only liquid Treasury-collateralised target on Mantle

## CRON

### GitHub Actions hourly

~0.005 MNT / cycle gas → free-tier sustainable, public verifiable workflow log

## SOURCIFY

### 5 contracts verified

Judge-readable source on Mantle Explorer, no opaque proxies

## INNOVATION IN TECHNICAL APPROACH

- Three-model adversarial consensus with cross-provider diversity (Bedrock + Vertex AI)
- ERC-8004 reference implementation — auto-updating tokenURI per cycle
- Synrail-inspired Discipline Layer — TX proof + freshness + drift gates after every swap
- RWA Allocator dual-path: LLM-driven (Path A) + deterministic idle-parking (Path B)



WHO BUILT THIS AND WHAT'S NEXT

# A solo developer. A repeatable pattern.

## TEAM

## USBVadik

Solo developer · ex-Synrail (autonomous-agent verification framework, also OSS)

[github.com/USBVadik/TuringVault-Core](https://github.com/USBVadik/TuringVault-Core)

[github.com/USBVadik/synrail](https://github.com/USBVadik/synrail)

[x.com/a\\_seven\\_life](https://x.com/a_seven_life)

## STACK DISCIPLINE

→ Spec-first workflow (8 specs, >500 acceptance criteria)

→ 156/156 unit tests · CI on every push

→ Honesty rule enforced as steering doc

→ Every deploy verifiable on Mantlescan + GitHub Actions log

## ROADMAP (POST-HACKATHON)

### Q3 2026

Vault contract for public deposits · DAO treasury onboarding · multi-asset RWA basket

### Q4 2026

Cross-agent reputation marketplace · ERC-8004 identity graph · per-decision IPFS streaming

### 2027

Open the Discipline Layer as an SDK · third-party agents stake reputation through TuringVault

## LIVE DEMO

[frontend-seven-beta-46.vercel.app](https://frontend-seven-beta-46.vercel.app)

## SOURCE

[github.com/USBVadik/TuringVault-Core](https://github.com/USBVadik/TuringVault-Core)

## CRON LOG

[Actions](#) / [Agent Cycle](#)

"The most valuable thing an AI RWA portfolio manager can do is **prove why it didn't allocate.**"